



First Implementation — Navigation App

This document describes all the implemented features in the navigation app and the general interaction flow for the user.

1. Feature Overview

The navigation app offers five integrated macro-features:

#	Feature	Description
TASK 1	Search with Places Autocomplete	Text search with real-time suggestions, debounce, and session token to optimize costs
TASK 2	Text search flow	Complete flow: typing → suggestions → selection → coordinates
TASK 3	Map point selection	Direct tap on the map to select a destination (POI or generic point)
TASK 4	Route calculation with alternatives	Route calculation from GPS position with alternative routes, selecting the fastest
TASK 5	Monitoring and recalculation	Continuous position check every 2s, switch to alternative or API recalculation if deviated
TASK 6	Stop Detection and Overlay	Shows turn instructions at waypoints or detects lateral roads when the vehicle stops (10s analysis)

2. File Structure

File	Role
<code>lib/services/places_service.dart</code>	Places API service: autocomplete + details + session token
<code>lib/services/directions_service.dart</code>	Directions API service: route calculation + alternative routes
<code>lib/services/roads_service.dart</code>	Roads API service: check for side roads

<code>lib/services/navigation_monitor.dart</code>	Business logic: position monitoring, bearing, deviation
<code>lib/services/geo_utils.dart</code>	Geometric functions: distance, point-on-polyline, thresholds
<code>lib/screens/navigation_screen.dart</code>	Main screen: coordinates UI and logic
<code>lib/widgets/search_input.dart</code>	Search widget with autocomplete and debounce
<code>lib/widgets/map_widget.dart</code>	Google Map with polyline, markers, and tap handler
<code>lib/widgets/directions_list.dart</code>	Step-by-step driving directions list
<code>lib/widgets/navigation_overlay.dart</code>	Navigation overlay: turn instructions, "go straight"

3. TASK 1 — Search with Places Autocomplete

3.1 Session Token

The session token is a unique identifier (UUID v4) that groups all queries of a single search session.

Token lifecycle:

1. User starts typing → **generates** a new token
2. Each Autocomplete request uses the **same** token
3. User selects a result → Places Details uses the **same** token
4. After selection → token is **invalidated**, a new one is generated

⚠ **Why it's important:** Without a session token, Google bills every single Autocomplete request separately. With the token, all Autocomplete + Details are grouped into a single billable session, **drastically reducing costs**.

3.2 Debounce (300ms)

A request is not sent for every typed character. A 300ms timer starts after each keystroke:

- If the user types another character before 300ms → the timer is **canceled and recreated**
- If the user stops typing for 300ms → the request is **sent**

Result: typing "Milan Cathedral" quickly sends ~2-3 requests instead of 15.

3.3 Minimum 3 Characters

No API request is sent if the typed text has fewer than 3 characters. Below this threshold, results would be too generic and would waste API calls with no benefit to the user.

4. TASK 2 — Complete Text Search Flow

- 1 User types in the search bar



- 2 If `input.length < 3` → no request (empty list)



- 3 If `input.length ≥ 3` → waits for the debounce (300ms)



- 4 Calls **Places Autocomplete API** with the session token



- 5 Shows the list of suggestions below the text field



- 6 User selects a suggestion



- 7 Calls **Places Details API** with the same session token → gets coordinates (lat/lng)



- 8 Ends the session → generates a **new** session token



- 9 Passes the coordinates to route calculation (TASK 4)

5. TASK 3 — Map Point Selection

Case 1 — Tap on a POI (shop, restaurant, registered place)

If the user taps a POI recognized by Google Maps, the map returns the POI coordinates. The coordinates are used directly to calculate the route, without additional API calls.

Case 2 — Tap on a generic point

If the user taps any point on the map that is not a POI:

- The lat/lng coordinates are obtained directly from the tap event
- **No call** to Places Details API (coordinates are sufficient)
- The destination text field shows the selected coordinates
- Proceeds directly to route calculation (TASK 4)

Optimization: No additional API is called for generic points. The lat/lng coordinates are sufficient for the Directions API.

6. TASK 4 — Route Calculation

After getting the destination (from text search or map selection), the system:

- 1 Gets the **current GPS position** of the device (used as origin)



- 2 Calls the **Directions API** with `alternatives: true`



- 3 Receives all routes (main + alternatives)



- 4 **Compares the duration** of each route and selects the fastest



- 5 **Saves all routes** with decoded polylines in memory



- 6 **Draws the active route *polyline*** on the map



- 7 Shows step-by-step driving directions in the list

Data models:

- `RouteData` — a single route (decoded polyline, duration in seconds, steps, distance)
- `AllRoutesResult` — complete result (bestRoute + list of all alternative routes)

7. TASK 5 — Position Monitoring and Automatic Recalculation

Once navigation starts, the system performs an automatic check **every 2 seconds**:

Step 2a — Is the user on the route?

Compares the current GPS position with the active route's polyline. If the minimum distance from the polyline is $\leq$ **40 meters**, the user is considered "on the route" → no action.

Step 2b — Is it on an alternative route?

If the user is more than 40m from the active route, the system checks if the current position is within 40m of the polyline of one of the **saved alternative routes** in memory:

- **If yes:** the alternative route becomes the new active route. **No API call** — the switch is instantaneous.
- **If no:** proceeds to Step 2c.

Step 2c — API Recalculation

If no alternative route is compatible, the system:

- Uses the **current GPS position as the new origin**
- Keeps the **same final destination**
- Calls the **Directions API with `alternatives: true`**
- Updates the active route and all alternatives
- Updates the map and directions

⚠ **Anti-flood:** An `_isRerouting` flag prevents concurrent API calls from being sent. If a recalculation is already in progress, subsequent timer ticks are ignored until completion.

8. TASK 6 — Stop Detection and Overlay

During navigation, the system constantly monitors speed to provide contextual instructions via a **Navigation Overlay**. When the speed drops below 1 km/h, a 10-second countdown activates to prevent false positives. When the timer expires, an analysis is performed:

Case A — Stopped at an intersection (Waypoint)

If the user is stopped within 5 meters of the route's next turn waypoint, the overlay appears showing the exact navigation instruction from the Directions API (e.g., "Turn right").

Case B — Lateral Roads Detection (Roads API)

If the user is stopped but is **NOT** on an expected intersection of the route, the system calculates lateral points relative to the vehicle's bearing and queries the **Roads API**. If a nearby lateral road is detected, a visual warning is shown to encourage the user to stay on the main route.

Overlay Dismissal: The overlay disappears automatically as soon as the user resumes driving and their speed exceeds the minimum threshold.

9. General Interaction Flow

The complete app flow, from destination selection to active navigation, develops as follows:

Phase	User Action	App Behavior
Entry	Opens the app	Shows map centered on Italy, starts GPS
Destination (text)	Types an address	300ms Debounce → Autocomplete → suggestions dropdown
Destination (map)	Taps a point on the map	Takes coordinates directly, no additional API
Selection	Chooses suggestion / confirms point	Places Details → coordinates → route calculation
Route	—	Directions API, selects fastest, draws polyline
Navigation	Follows the route	Monitoring every 2s, overlay with instructions

Deviation	Takes a different road	Switch to alternative or automatic API recalculation
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